Chronic Kidney Disease

Lori Wazny, BSc(Pharm), PharmD
Kaley Kawchuk, BSc(Pharm), PharmD
J. Christine Davis, BSc(Pharm), PharmD
Jay Hingwala, MSc, MD, FRCPC

Date of Revision: August 15, 2025

Peer Review Date: July 14, 2024

Introduction

Chronic kidney disease (CKD) is estimated to affect approximately 4 million Canadians.[1][2] It often coexists with cardiovascular disease and diabetes and is recognized as a risk factor for all-cause mortality and cardiovascular disease.[4]

Kidney function is described using the glomerular filtration rate (GFR) or creatinine clearance (ClCr). Estimated GFR (eGFR) is calculated and reported using the Modification of Diet in Renal Disease (MDRD) equation or the newer Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation.[5][6] The CKD-EPI equation was developed to provide an estimate of renal function that is closer to a measured value than the MDRD equation, especially when actual GFR is >60 mL/min/1.73m^2. An estimated creatinine clearance can also be calculated using the Cockcroft-Gault equation or measured using a 24-hour urine collection. The equations have limitations, especially at the extremes of age and muscle mass, but are more reliable than serum creatinine alone. See Dosage Adjustment in Renal Impairment and Renal Function Calculator.

An estimated or measured GFR of less than 60 mL/min/1.73 m^2 is considered abnormal for all adults. A value of more than 60 mL/min/1.73 m^2 is considered abnormal if it is accompanied by abnormalities of urine sediment or of imaging tests, or if the patient has had a kidney biopsy with abnormalities.

CKD is defined as the presence of kidney damage for a period greater than 3 months. The eGFRs used to characterize the stages of CKD are listed in Table 1.

***Table 1:** KDIGO Stages of Chronic Kidney Disease*[7]

KDIGO Stage	GFR (mL/min/1.73 m^2)	Description
G1	≥90	Normal or high
G2	60–89	Mildly decreased
G3a	45–59	Mildly to moderately decreased
G3b	30–44	Moderately to severely decreased
G4	15–29	Severely decreased
G5	<15	Kidney failure (add 'D' if treated by dialysis)

Abbreviations: GFR = glomerular filtration rate; KDIGO = Kidney Disease: Improving Global Outcomes

CKD was previously thought to be progressive, with the patient experiencing a decline in function over time and ultimately requiring dialysis. With the development of new interventions and prevention strategies, patients with CKD, particularly those without proteinuria, may have little progression of their kidney disease; however, they remain at high risk for cardiovascular events and death.[8] In patients with a preserved GFR, proteinuria alone is a risk factor for loss of kidney function (lower estimated GFR and higher albuminuria).[9]

This chapter summarizes the management of patients with chronic kidney disease excluding those on renal replacement therapy.

Goals of Therapy

- Slow the progression of CKD
- Manage reversible cardiovascular risk factors
- Treat the complications of CKD
- Educate the patient and caregivers on CKD

Investigations

- History:
  - many causes of CKD are hereditary, genetic or associated with other conditions, so both a patient and a family history are essential; risk factors for CKD are listed in Table 2 and patients with risk factors, especially cardiovascular disease and diabetes mellitus, should be screened for CKD[11]
  - a targeted review of symptoms related to the cause of kidney disease or its sequelae include shortness of breath, edema, pre-syncope, diarrhea, restless legs, pruritus, cramping, anorexia, nausea, fatigue, insomnia and hematuria; symptoms such as rashes, active joints, skin changes or hemoptysis may be suggestive of vasculitis or a collagen vascular disorder as a cause for the patient's CKD

Table 2: *Risk Factors for Chronic Kidney Disease*[7][10]

Age	Hypertension
Atherosclerotic vascular disease	Multiple myeloma
Autoimmune diseases, such as lupus, rheumatoid arthritis, connective tissue disease and vasculitis	Obesity
Chronic urinary tract obstruction from prostatic enlargement, neurogenic bladder, kidney stones	Pregnancy complications including edema, hypertension, proteinuria
Chronic viral infections, such as Hepatitis B and C, HIV	Recurrent episodes of dehydration
Diabetes mellitus	Recurrent pyelonephritis
First degree relative with CKD	Reduced nephron mass, e.g., congenital single kidney, post nephrectomy, scarring from reflux nephropathy
First Nations, Inuit, Metis or urban Indigenous people[a]	Tobacco dependence
Hereditary polycystic kidney disease	Use of known nephrotoxic drugs, e.g., NSAIDs including COX-2 inhibitors, lithium, acetaminophen (chronic use), cyclosporine, tacrolimus, contrast dyes
History of acute kidney injury	

[a] Chronic kidney disease risk is thought to be increased in this population from a combination of genetic factors and social, economic or health inequities.

- Physical exam:
  - general appearance: in advanced stages of CKD, patients develop cachexia or loss of muscle mass; advanced uremia is often accompanied by a sallow, greyish complexion and so-called "uremic fetor," which is urine-like breath odour secondary to breakdown of urea to ammonia in saliva
  - weight: measure at each visit to assess fluid and nutritional status; large changes are usually associated with fluid gains or loss
  - vitals: lying and standing blood pressures and pulse
  - cardiovascular: assessment of the jugular venous pressure, peripheral pulses, vascular bruits and presence of edema
  - respiratory: presence of adventitious sounds or decreased air entry can be related to hypervolemia
  - abdominal exam: palpation for enlarged, cystic kidneys, and auscultation for renal artery bruits at a position approximately 2 cm above and 2 cm lateral to the umbilicus; among patients with known hypertension, a bruit audible in both systole and diastole is moderately specific for renal artery stenosis[12]
- Laboratory investigations:
  - urine tests (see Table 3, Table 4)
    - urinalysis: may be included as an investigation to screen for hematuria (see Figure 1). Important limitations include its insensitivity to other urinary proteins apart from albumin and its inability to detect microalbuminuria (usually <2.5 mg/mmol). Urine dipstick results are also affected by hydration status. Urinalysis should be done in the following individuals:
      - patients at risk for CKD (see Table 2)
      - patients with abnormal creatinine
      - patients with diabetes at the time of diagnosis and yearly thereafter
      - patients with hypertension
    - urine albumin-to-creatinine ratio (ACR) or urine protein-to-creatinine ratio (PCR): these tests detect and quantify small amounts of proteinuria not detected by the urinalysis. The level of proteinuria is used to risk-stratify patients for cardiovascular and kidney events. Important limitations include transient (usually within 24 h) higher values from exercise, fever, poor glycemic control and inotropic medications; therefore, repetitive measures are more reliable
    - 24-hour urine collection: measure creatinine clearance or confirm albumin excretion rate (AER), protein excretion rate (PER) or other metabolic parameters; important limitations include reliable collection volumes, and transport and completion of the test being cumbersome for patients
  - blood tests (see Table 4 for suggested frequency)
    - CBC, Na, K, Cl, HCO_3, urea, creatinine (calculated eGFR), lipid profile
      - if diabetes present, add HbA_{1c}; initial investigations may also include serum protein electrophoresis and serum free light chain ratio[13]
      - at stage G3b or higher, add albumin, total calcium, phosphorus, parathyroid hormone, transferrin saturation, ferritin
      - if a glomerulonephritis is suspected clinically from symptoms such as active joints, rashes, or there is clinical or biochemical evidence of end organ damage such as significant proteinuria or active urine sediment, then possible tests include antinuclear antibody (ANA), antineutrophil cytoplasmic antibodies (ANCA, MPO and PR3 antibodies), anti-PLA2R antibody, cryoglobulins, hepatitis serology, HIV serology, serum C3 and C4, and rheumatoid factor (RF)

- imaging
  - renal ultrasound in patients who have increased creatinine, proteinuria or abnormal urine sediment to determine renal size and look for anatomic abnormalities, such as a solitary, small or polycystic kidney
  - renal biopsies can be performed to aid in diagnosis, prognostication or treatment decisions

Table 3: *Categories of Albuminuria and Tests Used to Quantify Them*[7]

Urine Test	Albuminuria Category		
	A1 (normoalbuminuria)	A2 (microalbuminuria)	A3 (macroalbuminuria)[a]
AER (mg/24 h)[b]	<30	30–300	>300
ACR (mg/mmol)	<3	3–30	>30
Protein reagent strip	Negative to trace	Trace to +	+ or greater

[a] Includes nephrotic syndrome (defined as albumin excretion >2200 mg/24 h or protein excretion >3000 mg/24 h).
[b] Requires a 24-hour urine collection.
Abbreviations: ACR = albumin-to-creatinine ratio; AER = albumin excretion rate

Table 4: *Recommended Frequency of Blood and Urine Screening Tests in Chronic Kidney Disease*[7]

	Testing Frequency (months)[a]		
KDIGO Stage	A1 albuminuria (normoalbuminuria)	A2 albuminuria (microalbuminuria)	A3 albuminuria (macroalbuminuria)
G1	12	12	6
G2	12	12	6
G3a	12	6	4
G3b	6	4	4
G4	4	4	2–3
G5	1–3	1–3	1–3

[a] See text for suggested blood tests. Urine tests include ACR (or PCR if indicated) and standard urinalysis. Urine culture and sensitivity only if urinary tract infection suspected.
Abbreviations: ACR = albumin-to-creatinine ratio; KDIGO = Kidney Disease: Improving Global Outcomes; PCR = protein-to-creatinine ratio

Patients may be referred to a nephrologist for the following conditions/results:[14]

- Acute kidney injury (≥26.5 μmol/L increase in creatinine within 48 h *or* 1.5 times over baseline increase in creatinine over 7 days *or* urine output <0.5 mL/kg/h over 6 h)
- eGFR <30 mL/min/1.73 m^2
- Progressive loss of kidney function, e.g., an eGFR <60 mL/min/1.73 m^2 that declines by ≥5 mL/min/1.73 m^2 in 6 months
- Hematuria or suspected glomerulonephritis
- Structural disease or polycystic kidney disease
- Persistent significant proteinuria (present on 2 out of 3 samples)
  - on dipstick *or* quantified PCR >100 mg/mmol *or* quantified ACR >60 mg/mmol
- Inability to achieve BP treatment targets or other difficulties in the management of the CKD patient

For patients with a new finding of eGFR between 30 and 60 mL/min/1.73 m^2, the physician should determine the stability of the patient's eGFR, repeat test within 2–4 weeks and then again in 3–6 months.[3] Consider reversible causes, such as intercurrent illness, volume depletion, medications (e.g., NSAIDs, aminoglycosides, IV contrast dye) and obstruction. If the eGFR remains between 30 and 60 mL/min/1.73 m^2, consider referral to a nephrologist.

The Ontario Renal Network developed the KidneyWise Clinical Toolkit to outline the identification and management for CKD in primary care. Included in the toolkit is the Kidney Failure Risk Equation, which may be used to estimate the risk of kidney failure in a patient with stages 3–5 CKD.[15]

Therapeutic Choices

Nonpharmacologic Choices[3][7]

Encourage patients to exercise for 30–60 minutes 4–7 days per week to reduce the possibility of becoming hypertensive or to lower blood pressure in those with hypertension.

Encourage smoking cessation to slow progression of CKD and to reduce the risk of cardiovascular disease (see Tobacco Use Disorder: Smoking and Nicotine Cessation).

Alcohol intake should be limited to 2 drinks or less per day as there is no known safe amount that will not elevate blood pressure.[16]

Patients with CKD and hypertension should follow a low sodium diet: <90 mmol/day Na, or 2 g Na or 5 g NaCl/day (see Special Diets).

If serum potassium >5 mmol/L, first consider medications that can be discontinued, such as potassium supplements, potassium-sparing diuretics and NSAIDs. If these medications are not present, advise dietary potassium restriction or consider increasing potassium excretion with either a diuretic (loop or thiazide/thiazide-like) or potassium binders. Note that angiotensin-converting enzyme inhibitors (ACEIs), angiotensin II receptor blockers (ARBs) and aldosterone antagonists may also contribute to the hyperkalemia; continue these agents if they're being used to decrease proteinuria and restrict dietary potassium instead, or increase potassium excretion from the body (see Potassium Disturbances).

Dietary protein intake has been the focus of several trials; however, there is a lack of convincing evidence that a long-term protein-restricted diet (<0.7 g/kg/day) delays the progression of CKD. A diet that is protein-controlled (0.8–1 g/kg/day) is recommended.

Referral to a renal dietitian is recommended to guide sodium, potassium, acidosis management and protein intake.

Encourage weight loss if patient has obesity (BMI >30 kg/m^2) or is overweight (BMI 25–29 kg/m^2) to lower the risk of CVD.

Pharmacologic Choices

Antihypertensives

Encourage patients with hypertension to purchase a Hypertension Canada–approved BP monitor for home use; suggest a monitoring plan and review these measurements at the medical follow-up. Blood pressure targets for nondiabetic CKD patients should be individualized (see Table 5). In adult patients with diabetes mellitus, the BP target is ≤130/80 mm Hg.[16]

Based on the results of the Systolic Blood Pressure Intervention Trial (SPRINT), a SBP target <120 mm Hg may be appropriate in patients 50–75 years of age with a 10-year Framingham risk >15% who do not have comorbidities such as diabetes, stroke or eGFR <60 mL/min/1.73 m^2.[18] This target also may be appropriate in patients who can achieve it without requiring a high number of antihypertensives and who are not experiencing adverse effects of therapy.

Patients prescribed ACEI/ARBs or diuretics who are targeting a SBP <120 mm Hg must be reliable and able to hold these medications on sick days when they are unable to maintain adequate fluid intake due to the risk of acute kidney injury (AKI).

***Table 5:** Individualized Blood Pressure Targets for Nondiabetic Chronic Kidney Disease*[16][17]

Patient Type	Systolic Blood Pressure Target
Meet SPRINT criteria[a]	<120 mmHg
Polycystic kidney disease	<110 mmHg[b]
Frail individuals, long-term care residents, previous stroke, limited life expectancy (<3 y), polypharmacy (>5 meds) and standing systolic BP <110	<140 mmHg
Note: Use caution when treating systolic BP to target; risks may outweigh benefits when diastolic BP <60 mm Hg	

[a] SPRINT criteria: patients >50 years of age, at elevated cardiovascular risk with systolic BP 130–180 mm Hg without diabetes, stroke or advanced CKD.
[b] For patients age 18-49 with BP >130/85 mmHg, and stage G1 or G2 as tolerated. In patients age >50, target systolic <120 mmHg as tolerated.

Table 6 presents guidelines on the choice of antihypertensive agent in CKD.

***Table 6:** Choices of Antihypertensive Medication in Chronic Kidney Disease*[16][19]

Clinical Condition	Antihypertensive
Diabetic kidney disease (ACR >3 mg/mmol)	ACEI or ARB as initial therapy. For persons in whom combination therapy with an ACEI is being considered, a dihydropyridine CCB is preferable to a thiazide/thiazide-like diuretic, based on the results of the ACCOMPLISH trial (see Hypertension).
Nondiabetic proteinuric CKD (ACR ≥30 mg/mmol or urine protein >500 mg/24 h)	ACEI or ARB. Thiazide/thiazide-like diuretics are recommended as additive antihypertensive therapy. For patients with CKD and volume overload, loop diuretics are an alternative.
Nondiabetic nonproteinuric CKD (ACR <30 mg/mmol)	Choose agents based on current hypertension guidelines (see Hypertension).

Abbreviations: ACEI = angiotensin-converting enzyme inhibitor; ACR = albumin-to-creatinine ratio; ARB = angiotensin receptor blocker; CCB = calcium channel blocker; CKD = chronic kidney disease

ACE Inhibitors and Angiotensin Receptor Blockers

ACE inhibitors (ACEIs) and **angiotensin II receptor blockers** (ARBs) are the preferred agents for certain types of CKD (see Table 6) because they have the following class effects:[19]

- Reduction of BP and intraglomerular pressure: in controlled trials, the beneficial effect of ACEIs and ARBs on the progression of CKD was greater than would be expected based on their antihypertensive effects alone.
- Reduction of proteinuria: ACEIs and ARBs reduce proteinuria more than any other antihypertensive, even when the effect of BP reduction on urinary protein excretion has been taken into account.
- Other mechanisms: ACEIs and ARBs alter the function of mesangial cells and interfere with angiotensin-mediated generation of free radical formation, which also helps slow the progression of CKD.

Increase the ACEI or ARB dose in patients whose BP is above target, and consider increasing in patients with higher levels of proteinuria if BP is within target, although careful follow-up of kidney function and BP will be required.[19] Moderate to high doses of ACEIs and ARBs have been associated with slowing the progression of CKD.[20] Begin with low doses and increase at 4- to 6-week intervals while monitoring for adverse effects and hypotension. The dosage range for each ACEI can be found in Hypertension, Table 7.

Measure eGFR and serum K^+ prior to and 2 weeks after initiating or increasing the dose of ACEI or ARB. Repeat ACR or PCR in 4–6 weeks.[19] Suggestions for monitoring and modifying ACEI and ARB therapy are provided in Table 7.

ACEIs/ARBs are contraindicated in pregnancy. Counsel premenopausal patients on appropriate contraception (see Contraception).

ACEIs/ARBs should be held if a patient has severe vomiting, diarrhea or volume depletion, and patients should be made aware of this.[21]

***Table 7:** Monitoring ACEI and ARB Therapy in Chronic Kidney Disease*[19]

Test	Monitoring Frequency	Action
eGFR	If eGFR ≥60 mL/min, repeat in 4–12 wk If eGFR 30–59 mL/min, repeat in 2–4 wk If eGFR <30 mL/min, repeat in ≤2 wk	Dosage adjustment is based on change in eGFR since previous test: • eGFR decreased by 0–14%: no dose change • eGFR decreased by 15–29%: no dose change, but repeat eGFR in 10–14 days • eGFR decreased by 30–50%: reduce dose and repeat eGFR every 5–7 days until eGFR within 30% of baseline • eGFR decreased by >50%: discontinue ACEI or ARB and repeat eGFR every 5–7 days until eGFR is within 15% of baseline value
K^+	If eGFR ≥60 mL/min, repeat in 4–12 wk If eGFR 30–59 mL/min, repeat in 2–4 wk If eGFR <30 mL/min, repeat in ≤2 wk	• K^+ 5–6 mmol/L: advise dietary potassium restriction • K^+ 6–6.5 mmol/L: prescribe loop diuretic if tolerated ± cation exchange resin

Abbreviations: ACEI = angiotensin-converting-enzyme inhibitor; ARB = angiotensin receptor blocker; eGFR = estimated glomerular filtration rate; K+ = serum potassium

Sodium-Glucose Cotransporter-2 Inhibitors

Sodium-glucose cotransporter-2 (SGLT2) inhibitors are now considered part of standard care in type 2 diabetic CKD.[7] They are also recommended for adults with CKD and albuminuria (urine ACR ≥20 mg/mmol) or those with heart failure (regardless of albuminuria).[7] Studies have shown SGLT2 inhibitors significantly slow progression of kidney disease, reduce the need for dialysis or transplantation, and decrease mortality in patients with and without diabetes.[7][22][23] When used for patients with type 2 diabetes, the degree of blood glucose lowering diminishes as GFR declines, but the beneficial renal effects are retained (see Figure 3). More recent studies used a lower eGFR cut-off of ≥20 mL/min/1.73 m^2 for starting SGLT2 inhibitors and continued until patients reached dialysis or kidney transplantation.[7][22]

SGLT2 inhibitors can cause an osmotic diuresis, which can lead to hypovolemia. The volume status of patients should be assessed prior to starting these drugs, and concurrent diuretics may require adjustment. An initial decline of 3–4 mL/min in eGFR is expected; if eGFR declines >20–25%, the medication should be stopped and the patient assessed for acute kidney injury (AKI).[24] The greatest risk of SGLT2 inhibitor use is euglycemic diabetic ketoacidosis (DKA). In adult patients with type 2 diabetes mellitus the risk is estimated to be 0.3–6.3 per 1000 person-years.[25][26] Most patients presenting with euglycemic DKA while taking an SGLT2 inhibitor will do so within 2 months of starting the medication. As well, most patients have a precipitating event, with dehydration, infection, surgery and changes in insulin dose being commonly reported. The incidence of DKA with SGLT2 inhibitors is significantly higher in patients with type 1 diabetes; use in these patients is not recommended.[27] More common adverse effects are genital mycotic infections and urinary tract infections. Keeping the perineal area clean and avoiding use in high-risk individuals, such as those with an indwelling urinary catheter, are risk-reduction strategies.[28]

Figure 3 suggests a process for initiating an SGLT2 inhibitor and shows the HbA_{1c} reductions that may be expected.

Diabetic Chronic Kidney Disease

Follow Diabetes Canada guidelines for glycemic targets (see Diabetes Mellitus).

First-line antihyperglycemic therapies for type 2 diabetes and CKD include metformin plus an SGLT2 inhibitor (see Figure 2). A figure outlining the comprehensive management of patients with diabetes and CKD is found in the online 2022 KDIGO Diabetes Guideline.

Discontinue **metformin** when ClCr or eGFR <30 mL/min/1.73 m^2 due to an increased risk of lactic acidosis.[29] Metformin should also be discontinued when there are acute decreases in kidney function, and illnesses or procedures that could lead to acute kidney injury (e.g., nausea/vomiting, dehydration, administration of IV contrast dye) or cause hypoxia (e.g., cardiac or respiratory failure), as these also are risk factors for lactic acidosis.[3] Diabetes Canada suggests medications (see Diabetes Mellitus - Chronic Complications, Table 1), including metformin, that should be held during acute illness.[30]

Patients with CKD are at higher risk of developing hypoglycemia because the ability of the kidney to metabolize insulin is impaired. In patients with eGFR <30 mL/min/1.73 m^2, emphasize how to recognize and treat hypoglycemia.

Glucagon-Like Peptide-1 Receptor Agonists

Glucagon-like peptide-1 (GLP-1) receptor agonists have been shown to improve glycemic control, reduce the risk of cardiovascular events and slow the progression of kidney disease in patients with type 2 diabetes. They are recommended in addition to metformin and an SGLT2 inhibitor for additional glucose lowering or as an alternative if these medications are not tolerated (see Figure 2).

The FLOW trial published in 2024 was a large (n = 3533) randomized, double blind, placebo-controlled trial comparing **semaglutide** 1 mg SC once weekly versus placebo in adults with type 2 diabetes and diagnosed CKD (eGFR 50–75 mL/min/1.73 m^2 and ACR 34–565 mg/mmol, or eGFR 25–49 mL/min/1.73 m^2 and ACR 11-565 mg/mmol).[31] Major kidney disease events and the annual rate of eGFR decline was reduced along with a reduction in major cardiovascular events and all-cause mortality with semaglutide.[31][32] Over 95% of patients were on a stable, maximally tolerated dose of renin-angiotensin system inhibitor; however, only about 16% of patients were on an SGLT2 inhibitor, and finerenone use was not reported. Although the benefit of combination therapy is theorized to be additive due to their complementary mechanisms of action, this has yet to be studied in a large outcome trial. Randomized trials assessing the efficacy and safety of combination therapy are anticipated.

The most common side effects associated with semaglutide are nausea, diarrhea and vomiting. Hypoglycemia may also occur, especially when used in combination with insulin. Patients with uncontrolled and potentially unstable diabetic retinopathy or maculopathy were excluded from the trial, as semaglutide has been associated with increased incidence of diabetic retinopathy complications.[31]

Non-steroidal Mineralocorticoid Receptor Antagonists

Finerenone is the first non-steroidal mineralocorticoid receptor antagonist (nsMRA) approved in Canada. It is indicated in patients with type 2 diabetic kidney disease, as an adjunct to standard therapy to reduce the risk of end-stage kidney disease and sustained decrease in eGFR, cardiovascular death, nonfatal myocardial infarctions, and hospitalization for heart failure. Two large randomized placebo-controlled trials in adult patients with CKD and type 2 diabetes demonstrated a reduction in the risk of CKD progression and cardiovascular events when added to maximum tolerated ACEI or ARB.[33][34][35][36] Very few patients were on SGLT2 inhibitors since the trials began before they became standard of care.

Finerenone is recommended in addition to maximum tolerated doses of ACEI or ARB and SGLT2 inhibitor in patients with type 2 diabetic kidney disease, an eGFR ≥25 mL/min/1.73 m^2, albuminuria (UACR ≥3 mg/mmol) and serum K ≤5 mmol/L. Finerenone may also be considered in patients on ACEI or ARB with albuminuria (UACR ≥3 mg/mmol) who do not tolerate an SGLT2 inhibitor.[36] Since clinical experience is limited, the manufacturer recommends discontinuing treatment when the eGFR falls below 15 mL/min/1.73 m^2.[37] Finerenone is contraindicated in combination with strong CYP3A4 inhibitors (e.g., itraconazole, ketoconazole, voriconazole, ritonavir, clarithromycin) and should not be used in combination with other mineralocorticoid antagonists such as spironolactone or eplerenone.[37]

Finerenone can cause hyperkalemia; therefore, routine monitoring of serum potassium is required. Dose adjustments as outlined in the drug monograph may be needed based on the test results.[7][37]

Cardiovascular Risk Reduction for Patients with Chronic Kidney Disease

Both a reduced eGFR and proteinuria confer substantial increases in progression of CKD, cardiovascular risk and death.[4][8] The prognosis associated with a given level of eGFR varies substantially based on the presence and severity of proteinuria. In fact, patients with heavy proteinuria but with overtly normal eGFR appear to have worse clinical outcomes than those with moderately reduced eGFR without proteinuria.[4]

Lipid-Lowering Agents for Cardiovascular Risk Reduction

Use a statin in patients with CKD who are ≥50 years of age or in those ≥18 years of age with CKD and diabetes, known coronary artery disease, prior stroke or 10-year Framingham risk >10%.[10] Suggested doses for primary prevention are presented in Table 8.

In adults with CKD and hypertriglyceridemia, therapeutic lifestyle changes are suggested. Treatment with fibrates is not recommended.[29]

Table 8: *Suggested Daily Doses of Lipid-Lowering Agents in Chronic Kidney Disease for Primary Prevention*[38]

Atorvastatin	20 mg	Pravastatin	40 mg	Simvastatin	40 mg
Fluvastatin	80 mg	Rosuvastatin	10 mg	Simvastatin/Ezetimibe	20 mg/10 mg

ASA for Primary Cardiovascular Risk Reduction

There is controversy over the use of **ASA** for primary prevention in patients with CKD. A meta-analysis reported no reductions in CV events or mortality, but major bleeding events were increased, which led to the conclusion that there is no clear benefit of ASA for primary prevention.[39] The randomized controlled trial data in this area is insufficient to recommend universal use or avoidance of ASA for primary prevention. Current CKD guidelines recommend ASA for patients with known ischemic cardiovascular disease (secondary prevention).[7]

Drug Therapy Adjustment in Patients with Chronic Kidney Disease

Dosage adjustment of drugs in renal impairment is described in Dosage Adjustment in Renal Impairment.

Very few drugs are absolutely contraindicated in patients with CKD; however, medications that are generally avoided in patients with stages G4–G5 CKD are listed in Table 9.

Table 9: *Examples of Medications to be Avoided in Stages G4–G5 Chronic Kidney Disease*

Medication	Complication
Aluminum-containing medications, e.g., antacids	For short-term use only (≤4 wk) in patients with severe hyperphosphatemia (serum phosphorus >2.23 mmol/L) despite treatment with other nonaluminum phosphate binders. Long-term use associated with aluminum toxicity (e.g., bone disease, dementia).
Apixaban	Caution with ClCr 15–24 mL/min. Limited data for stroke prevention in nonvalvular atrial fibrillation when ClCr <15 mL/min.
Baclofen	Increased neurotoxicity even at very low doses.[40]
Dabigatran	Not recommended with ClCr <30 mL/min due to increased risk of bleeding. In patients with ClCr 15–29 mL/min, 75 mg BID is suggested, but safety has not been established.[41][42]
Edoxaban	Patients with ClCr <30 mL/min were excluded from the venous thromboembolism trials. Nonvalvular atrial fibrillation: avoid if ClCr <15 mL/min.
Magnesium-containing medications, e.g., antacids, bowel preparations, laxatives	Magnesium accumulation.
Meperidine (pethidine)	Accumulation of an active metabolite that can lead to seizures.

Medication	Complication
Metformin	Risk of lactic acidosis with eGFR <30 mL/min/1.73m^2.
Morphine	Accumulation of active metabolites (morphine-3-glucuronide [CNS stimulation/lower seizure threshold], morphine-6-glucuronide [analgesic]) significant in patients with reduced kidney function. Use with caution and at reduced doses. Hydromorphone is preferred.
Nitrofurantoin	ClCr <30 mL/min: avoid use as ineffective (not excreted into the urine).
NSAIDs, COX-2 inhibitors and other nephrotoxins	Increased risk of acute kidney injury.
Phosphorus-containing products, e.g., phosphate enemas	Deaths due to hyperphosphatemia and resulting hypocalcemia have been reported; these products can also cause acute phosphate nephropathy.[43][44]
Potassium-sparing diuretics and herbals, e.g., alfalfa, dandelion, noni juice	Risk of hyperkalemia.[45]
Rivaroxaban	Caution with ClCr 15–30mL/min; not recommended <15 mL/min.
Sotalol	Risk of accumulation and torsades de pointes.[46]
Vitamin A	Risk of accumulation secondary to decreased renal catabolism and increased serum levels of retinol-binding protein.[47]
Vitamin C	Limit to no more than 60–100 mg/day as the metabolite (oxalate) can result in kidney stones and deposits of calcium oxalate in soft tissues.[47]

Complications of Chronic Kidney Disease

Complications are seen in stages G3–G5 CKD, and management by a nephrologist is recommended. These complications include:

Anemia: detailed clinical practice guidelines are provided by Kidney Disease: Improving Global Outcomes[48] (see also Common Anemias).

Hyperkalemia: refer to Potassium Disturbances.

Itchy skin: refer to Pruritus.

Metabolic acidosis: treat with **sodium bicarbonate** tablets or **Shohl solution** (citric acid/sodium citrate). Titrate the dose to achieve a CO_2 level ≥18 mmol/L.[7] Avoid serum bicarbonate >32 mmol/L as it is associated with increased mortality in patients with CKD.[49] Monitor closely since some patients will experience fluid retention and heart failure.

Mineral metabolism:

- If serum phosphate is progressively or persistently elevated despite a low phosphate diet (800–1000 mg phosphate/day), start therapy with a low dose of calcium-containing phosphate binder (**calcium carbonate** or **calcium citrate**) if hypercalcemia is not present.[3] See Table 4 for suggested testing frequency and Table 10 for dosing. The maximum safe dose of calcium-based phosphate binders with regards to vascular calcification is not known.
- If hypercalcemia develops, reduce the dose of calcium-containing phosphate binders and vitamin D analogues, if used. If hyperphosphatemia is still present, the patient may be changed to a non-calcium-containing phosphate binder, such as **lanthanum**, **sevelamer carbonate** or **sevelamer hydrochloride**. The carbonate salt of sevelamer may help to neutralize uremia-induced metabolic acidosis. The newest non-calcium-containing phosphate binder is **sucroferric oxyhydroxide**. Although it is iron-based, it contributes negligibly to total daily iron intake.
- The optimal serum intact parathyroid hormone (PTH) level is not known in CKD patients not receiving dialysis. Guidelines suggest that vitamin D analogues, such as **alfacalcidol** or **calcitriol**, not routinely be used, but reserved for patients with eGFR <30 mL/min/1.72 m^2 with severe and progressive hyperparathyroidism. These analogues are required since the kidney is less able to activate other forms of vitamin D. **Cinacalcet** is not recommended to treat hyperparathyroidism in nondialysis CKD patients due to an increased incidence of hypocalcemia.[50]

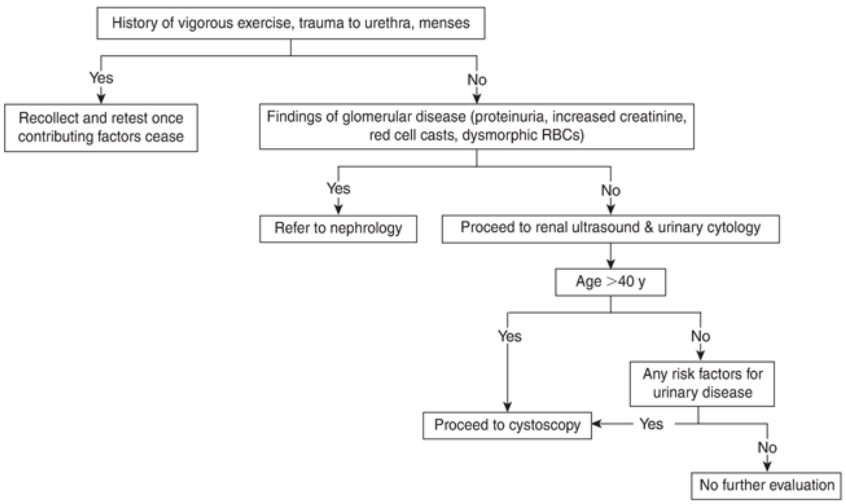
Therapeutic Tips

- Screen individuals at risk for CKD with an annual history and physical exam including BP assessment, calculation of eGFR, urinalysis, and spot urine for ACR or PCR.

- Start ACEIs or ARBs at moderate doses in those patients with normal GFR and titrate up to the maximally tolerated dose.
- SGLT2 inhibitors should be considered after ACEI and/or ARB for all patients with diabetic CKD to slow progression of kidney disease and reduce cardiovascular mortality and heart failure events. They may also be considered for use in nondiabetic CKD with ACR >20 mg/mmol.[23] Continue SGLT2 inhibitors until start of renal replacement therapy (dialysis or kidney transplant).
- Finerenone is recommended as an adjunct therapy with ACEI or ARB and SGLT2 inhibitor for patients with diabetic CKD and UACR ≥3 mg/mmol if eGFR ≥25 ml/min/1.73 m^2 and serum K ≤4.8 mmol/L.

Algorithms

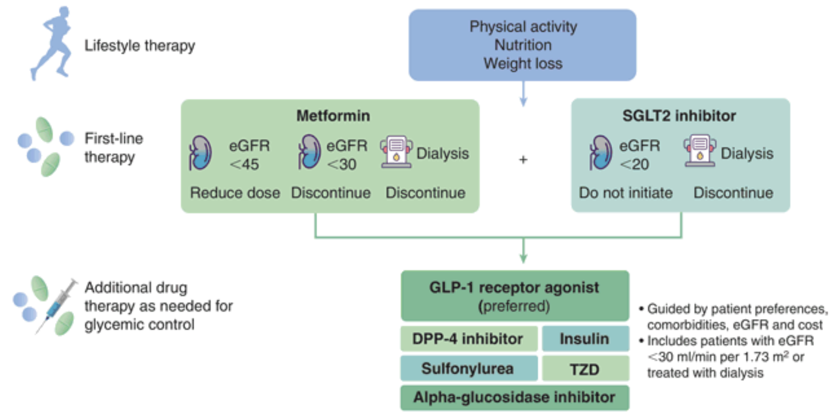
Figure 1: *Stepwise Investigation of Hematuria*



Abbreviations: RBC = red blood cell

Reproduced with permission from Wollin T, Laroche B, Psooy K. Canadian guidelines for the management of asymptomatic microscopic hematuria in adults. *Can Urol Assoc J* 2009;3(1):77-80.

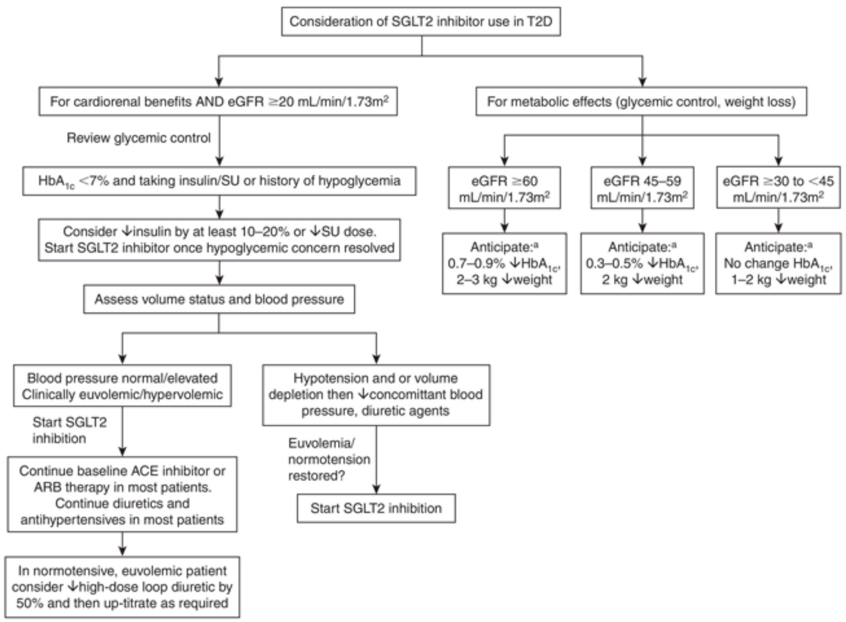
Figure 2: *Selecting Antihyperglycemic Drugs for Patients with Type 2 Diabetes and Chronic Kidney Disease*



Abbreviations: DPP–4 = dipeptidyl peptidase-4; eGFR = estimated glomerular filtration rate; GLP-1 = glucagon-like peptide-1; SGLT2 = sodium–glucose cotransporter-2; TZD = thiazolidinedione

Reprinted with permission from Elsevier. Kidney Disease: Improving Global Outcomes (KDIGO) Diabetes Work Group. KDIGO 2022 clinical practice guideline for diabetes management in chronic kidney disease. *Kidney Int* 102(5S):S1-S127. Copyright 2022.

Figure 3: *Suggested Process for Initiating an SGLT2 Inhibitor in Type 2 Diabetes*



[a] Also consider following guidance around adjustment of insulin and SU therapies, diuretics and antihypertensives prior to initiation of therapies.

1. *Note:* Check electrolytes, creatinine after initiation of treatment according to local guidelines/practice.

Abbreviations: ACE = angiotensin-converting enzyme; ARB = angiotensin II receptor blocker; eGFR = estimated glomerular filtration rate; SGLT2 = sodium-glucose cotransporter-2; SU = sulfonylurea; T2D = type 2 diabetes mellitus

Reprinted with permission from SAGE Publications, Inc. Dubrofsky L, Srivastava A, Cherney DZ. *Can J Kidney Health Dis* (7:2054358120935701). Copyright 2020 by Dubrofsky, Srivastava, Cherney.

Drug Table

Table 10: Drugs Used to Treat Chronic Kidney Disease

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions
Drug Class: Bicarbonate Supplements			
sodium bicarbonate generics <$50	Start at 325–500 mg BID or TID. Titrate to achieve an HCO_3^- level ≥18 mmol/L Max dose: 5850 mg/day 325 mg tablet = 3.8 mmol bicarbonate 500 mg tablet = 5.8 mmol bicarbonate Baking soda dissolved in water may be used as an alternative in patients who cannot take tablets (⅛ teaspoon = 7.1 mmol bicarbonate)	Bloating, flatulence, increased Na^+ absorption.	Reduced absorption of medications requiring an acidic gastric pH, e.g., atazanavir, calcium carbonate, iron tablets, itraconazole, ketoconazole.
citric acid/sodium citrate (Shohl solution) Dicitrate Solution, generics <$50	1 mmol bicarbonate/mL Start at 0.5 mmol/kg/day in 2–3 divided doses. Titrate to achieve an HCO_3^- level ≥18 mmol/L	Bloating, flatulence, increased Na^+ absorption.	Reduced absorption of medications requiring an acidic gastric pH, e.g., atazanavir, calcium carbonate, iron tablets, itraconazole, ketoconazole.
Drug Class: Calcimimetics			
cinacalcet generics ~$320	Start at 30 mg PO daily with food. Titrate every 2–4 wk to PTH <53 pmol/L	Diarrhea, nausea, vomiting, hypocalcemia, hypophosphatemia. Not for use in predialysis CKD.	Cinacalcet strongly inhibits CYP2D6 and can increase levels of metoprolol, flecainide, vinblastine, thioridazine and most tricyclic antidepressants.
Drug Class: Glucagon-Like Peptide-1 (GLP-1) Receptor Agonists			
semaglutide Ozempic	Initial: 0.25 mg weekly SC for 4 wk; increase	Nausea, vomiting, diarrhea, injection site	May reduce rate of absorption of some oral

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions
$200–250	to 0.5 mg weekly SC from wk 5 onward May increase to 1 mg weekly SC after a further 4 wk	reactions, acute pancreatitis (rare).	medications.
Drug Class: Non-steroidal Mineralocorticoid Receptor Antagonists			
finerenone Kerendia $100–150	If eGFR ≥25 to <60 mL/min/1.73m^2, start at 10 mg PO daily. May increase to 20 mg PO daily in 4 wk if eGFR has not decreased >30% If eGFR ≥60 mL/min/1.73m^2, start at 20 mg PO daily Hold if serum K^+ >5.5 mmol/L	Hyperkalemia, hyperuricemia and hypotension.	Contraindicated in combination with strong inhibitors of CYP3A4, e.g., itraconazole. Avoid combination with inducers of CYP3A4, e.g., phenobarbital, rifampin.
Drug Class: Phosphate Binders			
calcium carbonate Caltrate, Tums, generics <$50	Start at 250–500 mg elemental Ca^{++}/day TID PO with meals. Titrate to achieve a PO_4 level in the normal range Calcium carbonate is 40% elemental calcium	Constipation and nausea are the most common. Others: hypercalcemia.	Oral iron salts, fluoroquinolones, tetracyclines and levothyroxine: absorption reduced. Give 2 h before or 4 h after calcium. H_2-blockers (e.g., ranitidine), proton pump inhibitors and sodium bicarbonate increase gastric pH and reduce dissolution and phosphate binding of calcium carbonate.
calcium citrate generics <$50	Start at 300–900 mg elemental Ca^{++} (1–3 tablets) TID PO with meals. Titrate to achieve a PO_4 level in the normal range	Constipation and nausea are the most common. Others: hypercalcemia.	Oral iron salts, fluoroquinolones, tetracyclines and levothyroxine: absorption reduced. Give 2 h before or 4 h after calcium. H_2-blockers (e.g., ranitidine), proton pump inhibitors and sodium bicarbonate increase gastric pH and reduce dissolution and phosphate binding of calcium carbonate. Less dependent on acidic gastric pH for dissolution. May increase absorption of aluminum from aluminum-containing antacids.
lanthanum carbonate generics $100–150	Start at 250–500 mg TID PO with meals. May be used in combination with other PO_4 binders	Nausea, diarrhea, flatulence. Potential for accumulation of lanthanum due to GI absorption, but long-term clinical consequences unknown.	Reduced absorption of levothyroxine and mycophenolate mofetil. Administer lanthanum 2 h after these drugs.
sevelamer carbonate Renvela, generics $200–250	800–2400 mg TID PO with meals. May be used in combination with other PO_4 binders	Heartburn, bloating, gas.	Cholesterol-lowering drugs may need to be reduced, as sevelamer can lower LDL cholesterol by an average of 30%. Reduced absorption of ciprofloxacin, levothyroxine and mycophenolate mofetil. Administer sevelamer 2 h after these drugs.
sevelamer hydrochloride Renagel $200–250	800–2400 mg TID PO with meals. May be used in combination with other PO_4 binders	Heartburn, bloating, gas. If used alone in patients with nondialysis CKD, monitor CO_2 levels as can	Cholesterol-lowering drugs may need to be reduced, as sevelamer can lower LDL cholesterol by an average of 30%. Reduced absorption of

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions
		worsen uremic metabolic acidosis.	ciprofloxacin, levothyroxine and mycophenolate mofetil. Administer sevelamer 2 h after these drugs.
sucroferric oxyhydroxide Velphoro ~$400–$800	500–1000 mg TID PO with meals	Diarrhea, nausea, discoloration of stool.	Reduced absorption of other drugs. Administer alendronate and levothyroxine 1 h before sucroferric oxyhydroxide. Administer doxycycline 1 h before or 2 h after sucroferric oxyhydroxide.
Drug Class: Sodium-Glucose Cotransporter 2 Inhibitors			
canagliflozin Invokana $50–100	100 mg PO once daily	Volume depletion, urinary tract infections, genital mycotic infections. In those with T2DM: hypoglycemia, euglycemic diabetic ketoacidosis.	Reassess diuretic dose; consider dose reduction if euvolemic. Reassess sulfonylurea and insulin dose to avoid hypoglycemia in those with T2DM.
dapagliflozin Forxiga, generics <$50	10 mg PO once daily	Volume depletion, urinary tract infections, genital mycotic infections. In those with T2DM: hypoglycemia, euglycemic diabetic ketoacidosis.	Reassess diuretic dose; consider dose reduction if euvolemic. Reassess sulfonylurea and insulin dose to avoid hypoglycemia in those with T2DM.
empagliflozin Jardiance $50–100	10 mg PO once daily	Volume depletion, urinary tract infections, genital mycotic infections. In those with T2DM: hypoglycemia, euglycemic diabetic ketoacidosis.	Reassess diuretic dose; consider dose reduction if euvolemic. Reassess sulfonylurea and insulin dose to avoid hypoglycemia in those with T2DM.
Drug Class: Vitamin D Analogues			
alfacalcidol One-Alpha, generics <$50	Start at 0.25 mcg PO every other day or daily. Titrate to PTH 2–9 times the upper limit of normal	Hypercalcemia, hyperphosphatemia.	Phenytoin, carbamazepine, phenobarbital, thiazide diuretics may reduce levels of alfacalcidol.
calcitriol Rocaltrol, Calcitriol-Odan, other generics <$50	Start at 0.25 mcg PO every other day or daily. Titrate to PTH 2–9 times the upper limit of normal	Hypercalcemia, hyperphosphatemia.	Phenytoin, carbamazepine, phenobarbital, thiazide diuretics may reduce levels of calcitriol.

[a] Cost of 30-day supply of usual dose of drug, based on 70 kg weight; includes drug cost only.

Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.

Abbreviations: Ca = calcium; CKD = chronic kidney disease; CO_2 = carbon dioxide; eGFR = estimated glomerular filtration rate; GI = gastrointestinal; HCO_3^- = bicarbonate; LDL = low-density lipoprotein; Na = sodium; PO_4 = phosphate; PTH = parathyroid hormone; T2DM = type 2 diabetes mellitus

Suggested Readings

Kidney Disease: Improving Global Outcomes (KDIGO). CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int* 2024;105(4S):S117-S314. Available from: https://kdigo.org/guidelines/ckd-evaluation-and-management.

Matzke GR, Aronoff GR, Atkinson AJ et al. Drug dosing consideration in patients with acute and chronic kidney disease—a clinical update from Kidney Disease: Improving Global Outcomes (KDIGO). *Kidney Int* 2011;80(11):1122-37.

Rabi DM, McBrien KA, Sapir-Pichhadze R et al. Hypertension Canada's 2020 comprehensive guidelines for the prevention, diagnosis, risk assessment, and treatment of hypertension in adults and children. *Can J Cardiol* 2020;36(5):596-624.

References

1. Manns B, McKenzie SQ, Au F et al. The financial impact of advanced kidney disease on Canada Pension Plan and private disability insurance costs. *Can J Kidney Health Dis* 2017;4:2054358117703986.
2. Hill NR, Fatoba ST, Oke JL et al. Global prevalence of chronic kidney disease - a systematic review and meta-analysis. *PLoS One* 2016;11(7):e0158765.
3. Levin A, Hemmelgarn B, Culleton B et al. Guidelines for the management of chronic kidney disease. *CMAJ* 2008;179(11):1154-62.
4. Hemmelgarn BR, Manns BJ, Lloyd A et al. Relation between kidney function, proteinuria, and adverse outcomes. *JAMA* 2010;303(5):423-9.
5. Levey AS, Berg RL, Gassman JJ et al. Creatinine filtration, secretion and excretion during progressive renal disease. Modification of Diet in Renal Disease (MDRD) Study Group. *Kidney Int Suppl* 1989;27:S73-S80.
6. Levey AS, Stevens LA, Schmid CH et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med* 2009;150(9):604-12.
7. Kidney Disease: Improving Global Outcomes (KDIGO). CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int* 2024;105(4S):S117-S314. Available from: https://kdigo.org/guidelines/ckd-evaluation-and-management.
8. Go AS, Chertow GM, Fan D et al. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med* 2004;351(13):1296-305.
9. Gansevoort RT, Matsushita K, van der Velde M et al. Lower estimated GFR and higher albuminuria are associated with adverse kidney outcomes. A collaborative meta-analysis of general and high-risk population cohorts. *Kidney Int* 2011;80(1):93-104.
10. Cancer Care Ontario/Ontario Renal Network. (2020). *KidneyWise clinical toolkit* [PDF file]. Available from: www.ontariorenalnetwork.ca/sites/renalnetwork/files/assets/Clinical_Toolkit.pdf.
11. Fink HA, Ishani A, Taylor BC et al. Screening for, monitoring, and treatment of chronic kidney disease stages 1 to 3: a systematic review for the U.S. Preventive Services Task Force and for an American College of Physicians Clinical Practice Guideline. *Ann Intern Med* 2012;156(8):570-81.
12. Krijnen P, Steyerberg EW, Postma CT et al. Validation of a prediction rule for renal artery stenosis. *J Hypertens* 2005;23(8):1583-8.
13. Willrich MA, Katzmann JA. Laboratory testing requirements for diagnosis and follow-up of multiple myeloma and related plasma cell dyscrasias. *Clin Chem Lab Med* 2016;54(6):907-19.
14. Hingwala J, Wojciechowski P, Hiebert B et al. Risk-based triage for nephrology referrals using the kidney failure risk equation. *Can J Kidney Health Dis* 2017;4:2054358117722782.
15. Tangri N, Stevens LA, Griffith J et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA* 2011;305(15):1553-9.
16. Rabi DM, McBrien KA, Sapir-Pichhadze R et al. Hypertension Canada's 2020 comprehensive guidelines for the prevention, diagnosis, risk assessment, and treatment of hypertension in adults and children *Can J Cardiol* 2020;36(5):596-624.
17. Kidney Disease: Improving Global Outcomes (KDIGO) ADPKD Work Group. KDIGO 2025 clinical practice guideline for the evaluation, management, and treatment of autosomal dominant polycystic kidney disease (ADPKD). *Kidney Int* 2025;107(2S):S1-S239.
18. SPRINT Research Group, Wright JT, Williamson JD et al. A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med* 2015;373(22):2103-16.
19. Kidney Disease Outcomes Quality Initiative (K/DOQI). K/DOQI clinical practice guidelines on hypertension and antihypertensive agents in chronic kidney disease. *Am J Kidney Dis* 2004;43(5 Suppl 1):S1-S290.
20. Jafar JH, Stark PC, Schmid CH et al. Progression of chronic kidney disease: The role of blood pressure control, proteinuria, and angiotensin-converting enzyme inhibition a patient-level meta-analysis. *Ann Intern Med* 2003;139(4):244-52.
21. Schoolwerth AC, Sica DA, Ballermann BJ et al. Renal considerations in angiotensin converting enzyme inhibitor therapy: a statement for healthcare professionals from the Council on the Kidney in Cardiovascular Disease and the Council for High Blood Pressure Research of the American Heart Association. *Circulation* 2001;104(16):1985-91.
22. Herrington WG, Staplin N, Wanner C et al. Empagliflozin in patients with chronic kidney disease. *N Engl J Med* 2023;388(2):117-27.
23. Heerspink HJL, Stefánsson BV, Correa-Rotter R et al. Dapagliflozin in patients with chronic kidney disease. *N Engl J Med* 2020;383(15):1436-46.
24. Dubrofsky L, Srivastava A, Cherney DZ. Sodium-glucose cotransporter-2 inhibitors in nephrology practice: a narrative review. *Can J Kidney Health Dis* 2020;7:2054358120935701.
25. Al-Hindi B, Mohammed MA, Mangantig E et al. Prevalence of sodium-glucose transporter 2 inhibitor-associated diabetic ketoacidosis in real-world data: a systematic review and meta-analysis. *J Am Pharm Assoc (2003)* 2024;64(1):9-26.
26. Nuffield Department of Population Health Renal Studies Group; SGLT2 inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium. Impact of diabetes on the effects of sodium glucose co-transporter-2 inhibitors on kidney outcomes: collaborative meta-analysis of large placebo-controlled trials. *Lancet* 2022;400(10365):1788-1801.
27. Diaz-Ramos A, Eilbert W, Marquez D. Euglycemic diabetic ketoacidosis associated with sodium-glucose cotransporter-2 inhibitor use: a case report and review of the literature. *Int J Emerg Med* 2019;12(1):27.
28. Li J, Albajrami O, Zhuo M et al. Decision algorithm for prescribing SGLT2 inhibitors and GLP-1 receptor agonists for diabetic kidney disease. *Clin J Am Soc Nephrol* 2020;15(11):1678-88.
29. Lipscombe L, Booth G, Butalia S et al. Diabetes Canada 2018 clinical practice guidelines for the prevention and management of diabetes in Canada: pharmacologic glycemic management of type 2 diabetes in adults. *Can J Diabetes* 2018;42(Suppl 1):S88-S103. Available from: https://guidelines.diabetes.ca/cpg/chapter13. Accessed April 30, 2021.

30. Diabetes Canada. Diabetes Canada 2018 clinical practice guidelines for the prevention and management of diabetes in Canada. Appendix 8. Sick-day medication list. *Can J Diabetes* 2018;42(Suppl 1):S316. Available from: www.diabetes.ca/DiabetesCanadaWebsite/media/Health-care-providers/2018%20Clinical%20Practice%20Guidelines/Appendix-8-sick-day-medication-list.pdf?ext=.pdf.
31. Perkovic V, Tuttle KR, Rossing P et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med* 2024;391(2):109-21.
32. Collister D, Pannu N. In T2DM with CKD, semaglutide reduced major kidney disease events at 3 y. *Ann Intern Med* 2024;177(9):JC98.
33. Bakris GL, Agarwal R, Anker SD et al. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med* 2020;383(23):2219-29.
34. Pitt B, Filippatos G, Agarwal R et al. Cardiovascular events with finerenone in kidney disease and type 2 diabetes. *N Engl J Med* 2021;385(24):2252-63.
35. Agarwal R, Filippatos G, Pitt B et al. Cardiovascular and kidney outcomes with finerenone in patients with type 2 diabetes and chronic kidney disease: the FIDELITY pooled analysis. *Eur Heart J* 2022;43(6):474-84.
36. CADTH. CADTH reimbursement recommendation: finerenone (Kerendia). *Canadian Journal of Health Technologies* 2023;3(3). Available from: www.cadth.ca/sites/default/files/DRR/2023/SR0737REC-Kerendia.pdf.
37. Health Canada. Drug Product Database. *Kerendia* (Bayer Inc.) [drug monograph]. 2026. Available from: www.canada.ca/en/health-canada/services/drugs-health-products/drug-products/drug-product-database.html.
38. Kidney Disease: Improving Global Outcomes (KDIGO). Lipid Work Group. KDIGO 2013 clinical practice guideline for lipid management in chronic kidney disease. *Kidney Int Suppl* 2013;3(3):259-305. Available from: https://kdigo.org/guidelines/lipids-in-ckd.
39. Major RW, Oozeerally I, Dawson S et al. Aspirin and cardiovascular primary prevention in non-endstage chronic kidney disease: a meta-analysis. *Atherosclerosis* 2016;251:177-82.
40. Chen KS, Bullard MJ, Chien YY et al. Baclofen in patients with severely impaired renal function. *Ann Pharmacother* 1997;31(11):1315-20.
41. U.S. National Library of Medicine, DailyMed. *PRADAXA - dabigatran etexilate mesylate capsule* [internet]. Available from: https://dailymed.nlm.nih.gov/dailymed/lookup.cfm?setid=5db7f199-8752-4d24-85f7-e34ca8f4d02e. Accessed May 17, 2021.
42. Moore TJ, Cohen MR, Mattison DR. Dabigatran, bleeding, and the regulators. *BMJ* 2014;349:g4517.
43. Piccoli GB, Vigotti FN, Consiglio V et al. Quiz page. Severe hypocalcemia caused by intravascular calcium phosphate precipitation after sodium phosphate-containing bowel preparation. *Am J Kidney Dis* 2010;55(2):A35-7.
44. U.S. Food and Drug Administration. *Information on oral sodium phosphate (OSP) products for bowel cleansing (marketed as Visicol and OsmoPrep, and oral sodium phosphate products available without a prescription)* [internet]. February 19, 2016. Available from: www.fda.gov/drugs/postmarket-drug-safety-information-patients-and-providers/information-oral-sodium-phosphate-osp-products-bowel-cleansing-marketed-visicol-and-osmoprep-and. Accessed May 12, 2021.
45. Isnard Bagnis C, Deray G, Baumelou A et al. Herbs and the kidney. *Am J Kidney Dis* 2004;44(1):1-11.
46. Dancey D, Wulffhart Z, McEwan P. Sotalol-induced torsades de pointes in patients with renal failure. *Can J Cardiol* 1997;13(1):55-8.
47. Rocco MV, Ikizler TA. Nutrition. In: Daugirdas JT, Blake PG, Ing TS, editors. *Handbook of dialysis*. 4th ed. Philadelphia (PA): Wolters Kluwer/Lippincott Williams & Wilkins; 2007. p. 462-81.
48. Kidney Disease: Improving Global Outcomes (KDIGO). Anemia Work Group. KDIGO clinical practice guideline for anemia in chronic kidney disease. *Kidney Int Suppl* 2012;2(4):279-335. Available from: https://kdigo.org/guidelines/anemia-in-ckd.
49. Navaneethan SD, Schold JD, Arrigain S et al. Serum bicarbonate and mortality in stage 3 and stage 4 chronic kidney disease. *Clin J Am Soc Nephrol* 2011;6(10):2395-402.
50. Health Canada. *Health Canada endorsed important safety information on Sensipar (cinacalcet hydrochloride)—for health professionals* [internet]. June 27, 2007. Available from: www.healthycanadians.gc.ca/recall-alert-rappel-avis/hc-sc/2007/14477a-eng.php. Accessed April 30, 2021.